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Question 1 [C01] [10 Points]

Write a java program to solve the following. A hotel has two room types: Studio and Apartment. The price per night depends on the month number (1 = January, 5 = May, etc.)

Month Number	Month Name	Studio(USD)	Apartment(USD)
5	May	50	65
6	June	75	69
7	July	76	77
8	August	76	77
9	September	75	69
10	October	50	65

Only these 6 months are valid. No need to handle any other month number.
If more than 14 nights:

Month(s)	Studio Discount	Apartment Discount
May or October (5/10)	30% off	10% off
June or September (6/9)	20% off	10% off
July or August (7/8)	No discount	10% off

Input	Output
Enter month: 5 Enter Total Nights: 15	Apartment: 877.5 USD. Studio: 525.0 USD.
Enter month: 8 Enter Total Nights: 13	Apartment: 1001.0 USD. Studio: 988.0 USD.

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Question 1 [C01] [10 Points]

Write a java program to solve the following. A beach resort offers Bungalow and Villa rooms. Price per night depends on month number (5 = May, ..., 10 = October):

Month	Bungalow (USD)	Villa (USD)
5 (May)	55	80
6 (June)	85	120
7 (July)	95	135
8 (August)	95	135
9 (September)	85	120
10 (October)	55	80

Only these 6 months are valid. No need to handle any other month number.
Extra Charge Rule (only if nights > 12)

Month(s)	Bungalow Extra	Villa Extra
5 or 10 (May/Oct)	+5 USD/night	+8 USD/night
6 or 9 (Jun/Sep)	+3 USD/night	+8 USD/night
7 or 8 (Jul/Aug)	No extra	+8 USD/night

Extra charge is added per night after base price.

Input	Output
Enter month: 5 Enter nights: 15	Villa: 1320.0 USD. Bungalow: 900.0 USD.
Enter month: 7 Enter nights: 10	Villa: 1350.0 USD. Bungalow: 950.0 USD.

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Question 1 [C01] [10 Points]

In the mystical kingdom of **Numoria**, scholars have uncovered an ancient **P-digit Number** etched into a crystal tablet. Your task is to decipher its power by examining the number digit by digit. Scholars believe this number holds hidden power, unlocked only by a special digit **N**.

As you explore the crystal, examine the number digit by digit:

- If the digit **N** is found in an odd position of the number, the magic is revealed as the sum of all the digits in odd positions.
- If **N** is found in an even position, the magic is revealed as the sum of all the digits in even positions.
- If **N** is not found anywhere in the number, then no magic is possible.

Write a java program to print whether **N** was found and the resulting magical sum (or indicate that no magic is possible).

Sample Input 1	Sample output 1
Ancient Number: 345689 Special digit: 5	Special digit 5 is found in the 4th position. Magical sum is : 16 Explanation: As the number is found in an even position from the right, magical sum = $8 + 5 + 3 = 16$
Sample input 2	Sample Output 2
Ancient Number: 93578 Special digit: 2	No Magic is possible

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Question 1 [C01] [10 Points]

In the enchanted land of **Eldoria**, a mysterious crystal has been discovered, inscribed with a **P-digit Ancient Number**. Scholars believe this number holds hidden power, unlocked only by a special digit **N**.

As you explore the crystal, examine the number digit by digit:

- If the special digit **N** is found in an odd position, the crystal awakens with magic equal to the product of all digits in odd positions.
- If **N** is found in an even position, the crystal's energy is revealed as the product of all digits in even positions.
- If **N** is not found, the crystal remains dormant, and no magic is possible.

Write a Java program to display whether **N** was found and the resulting magical product, or indicate that no magic is possible.

Sample Input 1	Sample output 1
Ancient Number: 345689 Special digit: 5	Special digit 5 is found in the 4th position. Magic number is : 120 Explanation: As the number is found in an even position from the right, magic number = $8 * 5 * 3 = 120$
Sample input 2	Sample Output 2
Ancient Number: 93578 Special digit: 2	No Magic is possible

[SET-A]

SOLUTION

```
import java.util.Scanner;
public class HotelPricing {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter month: ");
        int month = sc.nextInt();

        System.out.print("Enter Total Nights: ");
        int nights = sc.nextInt();

        // ---- Base nightly prices ----
        double studioNight = 0.0;
        double apartmentNight = 0.0;

        if (month == 5 || month == 10) {      // May or October
            studioNight = 50.0;
            apartmentNight = 65.0;
        } else if (month == 6 || month == 9) { // June or September
            studioNight = 75.0;
            apartmentNight = 69.0;
        } else if (month == 7 || month == 8) { // July or August
            studioNight = 76.0;
            apartmentNight = 77.0;
        }

        // ---- Full price ----
        double totalStudio = studioNight * nights;
        double totalApartment = apartmentNight * nights;

        // ---- Discounts (only when nights > 14) ----
        if (nights > 14) {
            if (month == 5 || month == 10) {
                totalStudio *= 0.70; // 30 % off
                totalApartment *= 0.90; // 10 % off
            } else if (month == 6 || month == 9) {
                totalStudio *= 0.80; // 20 % off
                totalApartment *= 0.90; // 10 % off
            } else if (month == 7 || month == 8) {
                totalApartment *= 0.90; // 10 % off only
            }
        }

        System.out.println("Apartment: " + totalApartment + " USD.");
        System.out.println("Studio: " + totalStudio + " USD.");

        sc.close();
    }
}
```

[SET-B]

SOLUTION

```
import java.util.Scanner;
public class BeachResort {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter month: ");
        int month = sc.nextInt();

        System.out.print("Enter nights: ");
        int nights = sc.nextInt();

        double bungalowBase = 0.0;
        double villaBase = 0.0;

        // Set base prices using if-else
        if (month == 5 || month == 10) {
            bungalowBase = 55.0;
            villaBase = 80.0;
        } else if (month == 6 || month == 9) {
            bungalowBase = 85.0;
            villaBase = 120.0;
        } else if (month == 7 || month == 8) {
            bungalowBase = 95.0;
            villaBase = 135.0;
        }

        // Calculate base total
        double totalBungalow = bungalowBase * nights;
        double totalVilla = villaBase * nights;

        // Apply extra charge only if nights > 12
        if (nights > 12) {
            double bungalowExtra = 0.0;
            double villaExtra = 0.0;

            if (month == 5 || month == 10) {
                bungalowExtra = 5.0;
                villaExtra = 8.0;
            } else if (month == 6 || month == 9) {
                bungalowExtra = 3.0;
                villaExtra = 8.0;
            } else if (month == 7 || month == 8) {
                villaExtra = 8.0;
            }

            totalBungalow += bungalowExtra * nights;
        }
    }
}
```

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```
        totalVilla += villaExtra * nights;
    }

    // Output: Villa first, then Bungalow
    System.out.println("Villa: " + totalVilla + " USD.");
    System.out.println("Bungalow: " + totalBungalow + " USD.");

    sc.close();
}
}
```

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In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A] SOLUTION

```
import java.util.Scanner;
public class Quiz2SetA {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Ancient Number: ");
        int num = sc.nextInt();
        System.out.print("Special digit: ");
        int N = sc.nextInt();
        int temp = num;
        int foundPos = -1;
        int temp_pos = 1;
        // Search the Number
        while (temp > 0) {
            int digit = temp % 10;
            if (digit == N) {
                foundPos = temp_pos;
                break;
            }
            temp /= 10;
            temp_pos++;
        }
        if (foundPos == -1) {
            // Number not found
            System.out.println("No Magic is possible");
        } else {
            // Number found
            System.out.println("Special digit " + N + " is found in the " + foundPos + "th position.");
            int sum = 0;
            temp = num;
            int i = 1;
            //loop for summation
            while (temp > 0) {
                int digit = temp % 10;
                if (foundPos % 2 == 0 && i % 2 == 0) {
                    sum += digit;
                } else if (foundPos % 2 != 0 && i % 2 != 0) {
                    sum -= digit;
                }
                temp /= 10;
                i++;
            }
            System.out.println("Magic number is : " + sum);
        }
    }
}
```


[SET-B]

SOLUTION

```
import java.util.Scanner;
public class Quiz2SetB {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Ancient Number: ");
        int num = sc.nextInt();
        System.out.print("Special digit: ");
        int N = sc.nextInt();

        int temp = num;
        int foundPos = -1;
        int temp_pos = 1;
        // Search the Number
        while (temp > 0) {
            int digit = temp % 10;
            if (digit == N) {
                foundPos = temp_pos;
                break;
            }
            temp /= 10;
            temp_pos++;
        }

        if (foundPos == -1) {
            // Number not found
            System.out.println("No Magic is possible");
        } else {
            // Number Found
            System.out.println("Special digit " + N + " is found in the " + foundPos
+ "th position.");
            int product = 1;
            temp = num;
            int i = 1;
            //loop for multiplication
            while (temp > 0) {
                int digit = temp % 10;
                if (foundPos % 2 == 0 && i % 2 == 0) {
                    product *= digit;
                } else if (foundPos % 2 != 0 && i % 2 != 0) {
                    product *= digit;
                }
                temp /= 10;
                i++;
            }
            System.out.println("Magic number is : " + product);
        }
    }
}
```